

ARCHITECTURE BASE PIPELINE

Live News / Market Feeds



Event Streaming Layer
(Kafka / Redpanda / NATS)



Multi Agent Signal Engine
- Technical Analysis



- Fundamental Analysis



- Sentiment Analysis



Confidence Arbitration Layer
(weighted ensemble scoring)



Strategy Execution Simulator
(latency + slippage aware)



Deterministic Replay Judge



Risk-adjusted Evaluation Engine



Leaderboard + Live Analytics Dashboard

Market Sentiment Analysis Parameters:

1. Provides a rating from -5 to +5 per segment.
2. Total Neural Network weight is 100%
3. Filters out the stocks based on the parameters.
4. High compute, resources and time cost metric.
5. Good for final validations and breakthrough of restrictions on current fundamental and technical analysis logics.
6. HOW THINGS WORK: Shortlisted stocks get a score between -5 to +5 (in respective parameters) and they're multiplied with the weight (of respective parameters). Stocks High-to-Low get arranged.
7. WHAT MAKES IT SPECIAL: Provides qualitative multi-dimensional understanding of perceived quantitative data.

#	Method	What It Measures
1	News Sentiment Analysis	Positive/negative tone in financial news articles.
2	Social Media Sentiment	Public opinion from X, Reddit, LinkedIn, Facebook, etc.
3	Search Trend Analysis	Rising interest using search volume data.
4	Options Market Sentiment	Expectations implied by options traders.
5	Institutional Fund Flow Analysis	Where large investors are moving capital.
6	Analyst Rating Sentiment	Upgrades, downgrades, and target-price revisions.
7	Earnings Call Sentiment	Management confidence and tone during earnings calls.
8	Insider Trading Analysis	Actions of executives and directors.
9	Technical Sentiment Indicators	Market psychology reflected in price behavior.
10	Consumer Review Sentiment	Customer opinions about products/services.
11	Supply Chain Sentiment	Signals from suppliers, logistics, and manufacturing.
12	Influencer, Community & Forum Analysis	Deep discussions from niche communities.

13	Macroeconomic Sentiment Analysis	Sentiment implied by economic indicators.
14	Alternative Data Sentiment	Real-world behavioral signals.
15	Prediction Market Analysis	What people are willing to bet on happening.

Technical Analysis Metrics:

1. Use relevant multi-dimensional perspective metrics for market stock filtration.
2. Can be used as trigger choices.
3. HOW THINGS WORK: Use multiple trigger initiators (group/ set of metrics) to pick diverse asset choices.
4. WHAT MAKES IT SPECIAL: Filters out the largest chunk to focus on a niche segment.

#	Metric	Measures
1	Trend Strength (ADX)	Whether a trend exists and how strong it is
2	Relative Strength vs Benchmark	Outperformance/underperformance
3	Momentum Rate of Change (ROC)	Speed of price movement
4	RSI Regime Analysis	Persistent buying/selling pressure
5	MACD Histogram Slope	Momentum acceleration/deceleration
6	Realized Volatility	Actual market uncertainty
7	ATR Expansion/Contraction	Volatility regime shifts
8	Bollinger Band Width	Compression vs expansion phases
9	Volume Profile	Price levels with highest participation
10	VWAP Distance	Institutional positioning

11	On-Balance Volume (OBV)	Whether volume confirms moves
12	Accumulation/Distribution Line	Smart money accumulation/distribution
13	Market Breadth	How many assets participate
14	Advance-Decline Line	Internal market health
15	New High/New Low Ratio	Leadership strength
16	Correlation to Market	Independence from index behavior
17	Beta Stability	Sensitivity to market movements
18	Drawdown Depth	Downside risk profile
19	Hurst Exponent	Trending vs mean-reverting nature
20	Market Structure Analysis	Higher highs/lows and trend integrity

Fundamental Analysis Metrics:

1. Use validation methods to support technical analysis.
2. Further filtering of choices.
3. HOW THINGS WORK: Validates stocks to validate choices.
4. WHAT MAKES IT SPECIAL: Doesn't follow market trend but verifies potential, consistency and reliability

#	Metric	What It Validates
1	Revenue Growth Rate	Demand growth
2	Organic Revenue Growth	True growth vs acquisitions
3	Gross Margin	Pricing power and moat
4	Operating Margin	Operational efficiency

5	Free Cash Flow Margin	Ability to convert revenue into cash
6	Return on Invested Capital (ROIC)	Capital efficiency
7	Return on Equity (ROE)	Shareholder value creation
8	Return on Assets (ROA)	Asset productivity
9	Free Cash Flow Growth	Sustainability of business expansion
10	Earnings Quality Ratio	Quality of reported earnings
11	Debt-to-EBITDA	Leverage risk
12	Interest Coverage Ratio	Debt servicing ability
13	Current Ratio	Short-term financial health
14	Share Dilution Rate	Management alignment
15	Insider Ownership Trend	Management conviction
16	Customer Concentration	Revenue risk
17	R&D Intensity	Future innovation pipeline
18	Revenue per Employee	Organizational efficiency
19	EV/FCF Ratio	Cash-based valuation
20	PEG Ratio	Whether growth justifies valuation